

## MDU 50X Modules



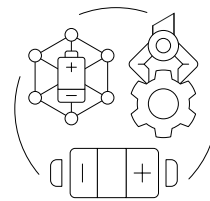
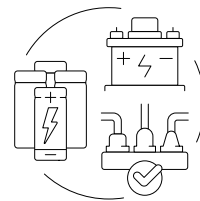
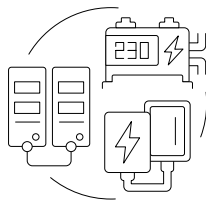
**50kVA to 600kVA Modular UPS**



**3 Phase Power System**



**Lithium Battery power System**



**LabPower-MDU 3P RM600/100X**

The LabPower-MDU 3P 50X modules, each rated at 50kW, can be combined to achieve capacities of 200kW, 300kW, 400kW, 500kW, and 600kW. This centralized bypass, 3-input-3-out modular UPS features digital control, a modular architecture, and N+X parallel redundancy technology. With flexible capacity options from 50kVA to 600kVA per module, its scalable design facilitates easy configuration and gradual investment. Supporting up to four parallel units, it delivers a total capacity of up to 2400kVA, making it ideal for high-demand, mission-critical environments.



### Applications

The LabPower-MDU product provides versatile power protection solutions across diverse sectors, ensuring safety, stability, and dependability. Its environmentally friendly design makes it suitable for critical applications such as data centers, financial institutions, government and corporate data rooms, healthcare facilities, educational institutions, and other environments that demand uninterrupted, high-quality power supply.



### Performance Features

#### Advanced Technology

- Digital control for reliable performance and high system stability
- Supports completely unbalanced loads
- Uses advanced three-level technology, achieving up to 96% efficiency
- Output power factor is 1

#### Convenient and flexible

- Modular design allows flexible capacity expansion with online addition or removal of modules
- Minimal downtime for quick maintenance and online repairs during emergencies
- Color touch display offers comprehensive UPS status, warnings, and fault information

#### High Reliability

- Redundant parallel central control unit ensures no system impact if one fails, with N+X redundancy achieving 99.999% availability
- Hot-swappable centralized monitoring module
- Uses centralized bypass power supply to enhance power capacity
- Equipped with EPO and remote emergency shutdown (REPO) functions

#### Intelligent Design

- LCD supports setting battery capacity and charging current, enabling automatic smooth switch of charging modes
- Features advanced two- and three-stage intelligent charging for fast charging and longer battery lifespan.



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## Specifications

| Models                             |                            | RM200/50X                                                                                                                                                                        | RM300/50X | RM400/50X     | RM500/50X | RM600/50X |
|------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|-----------|-----------|
| Nominal Capacity                   | Cabinet                    | 50~200kW                                                                                                                                                                         | 50~300kW  | 50~400kW      | 50~500kW  | 50~600kW  |
|                                    | Module                     | 50kW                                                                                                                                                                             |           |               |           |           |
| Maximum Number of Madoules         |                            | 4                                                                                                                                                                                | 6         | 8             | 10        | 12        |
| Input                              |                            |                                                                                                                                                                                  |           |               |           |           |
| Main Path Input                    | Nominal input voltage      | 380/400/415Vac                                                                                                                                                                   |           |               |           |           |
|                                    | input voltage range        | 138~485Vac; 305~485Vac no derating, 138~305Vac linear derating                                                                                                                   |           |               |           |           |
|                                    | Wiring system              | three phases and five wires (TN-S)                                                                                                                                               |           |               |           |           |
|                                    | Input frequency range      | 40-70Hz                                                                                                                                                                          |           |               |           |           |
|                                    | Input power factor         | ≥0.99                                                                                                                                                                            |           |               |           |           |
|                                    | Input THDi                 | ≤3% (100% linear load)                                                                                                                                                           |           |               |           |           |
| Bypass input                       | Nominal input voltage      | 380/400/415Vac                                                                                                                                                                   |           |               |           |           |
|                                    | Input voltage range        | 220Vac upper limit: 25% (optional+10%,+15%,+20%); 230Vac upper limit: 20% (optional+10%,+15%); 240Vac upper limit: 15% (optional+10%); lower limit: -45%(optional-10%,-20%,-30%) |           |               |           |           |
|                                    | Wiring system              | three phases and five wires (TN-S)                                                                                                                                               |           |               |           |           |
|                                    | Bypass frequency tolerance | ±10%                                                                                                                                                                             |           |               |           |           |
| Output                             |                            |                                                                                                                                                                                  |           |               |           |           |
| Voltage                            |                            | 380/400/415Vac±1%                                                                                                                                                                |           |               |           |           |
| Power factor                       |                            | 1                                                                                                                                                                                |           |               |           |           |
| Frequency(Hz)                      | Mains mode                 | ±1%/±2%/±4%/±5%/±10% settable                                                                                                                                                    |           |               |           |           |
|                                    | Battery mode               | (50/60±0.1%)Hz                                                                                                                                                                   |           |               |           |           |
| Waveform                           |                            | sine wave                                                                                                                                                                        |           |               |           |           |
| Peak current ratio                 |                            | 3:1                                                                                                                                                                              |           |               |           |           |
| Output THDV                        |                            | ≤2% (100% linear load); ≤3% (100% nonlinear load)                                                                                                                                |           |               |           |           |
| Switching time (ms)                |                            | Mains mode to bypass mode: 0ms; Mains mode to battery mode: 0ms                                                                                                                  |           |               |           |           |
| Efficiency (%)                     |                            | 95.50%                                                                                                                                                                           |           |               |           |           |
| Overload capacity                  |                            | 110% overload, continue for 60min turn to bypass mode; 125% overload, continue for 10min turn to bypass mode; 150% overload, continue for 1min turn to bypass mode               |           |               |           |           |
| Battery                            |                            |                                                                                                                                                                                  |           |               |           |           |
| Max charging current of the module |                            | 15A                                                                                                                                                                              |           |               |           |           |
| Battery voltage                    |                            | optional voltage: ±180V/±192V/±204V/±216V/±228V/±240/±252/±264/±276/±288/±300Vdc (30/32/34/36/38/40/42/44/46/48/50 sections, amount to 30~50 sections are optional)              |           |               |           |           |
| Environment                        |                            |                                                                                                                                                                                  |           |               |           |           |
| Operating temperature              |                            | 0°C~40°C                                                                                                                                                                         |           |               |           |           |
| Storage temperature                |                            | -25°C~55°C (exclude battery)                                                                                                                                                     |           |               |           |           |
| Humidity range                     |                            | 0~95% (no condensation)                                                                                                                                                          |           |               |           |           |
| Altitude of operation              |                            | < 1500m (More than 1500m is used in accordance with GB/T3859.2)                                                                                                                  |           |               |           |           |
| Noise (dB) (distance of one meter) |                            | <65dB                                                                                                                                                                            | <68dB     | <70dB         |           |           |
| Other functions                    |                            |                                                                                                                                                                                  |           |               |           |           |
| Warning                            |                            | Overload, Mains power anomaly, UPS fault, battery undervoltage, etc                                                                                                              |           |               |           |           |
| Protection                         |                            | Short circuit, overload, over temperature, battery under voltage, output under voltage, fan fault alarm, lightning protection, bypass back irrigation                            |           |               |           |           |
| Communication                      |                            | CAN, RS485, Network interface, dry contact, parallel computer interface, LBS interface, intelligent slot, temperature sensor interface                                           |           |               |           |           |
| Physical Informations              |                            |                                                                                                                                                                                  |           |               |           |           |
| Dimensions (D×W×H)(mm)             | Cabinet                    | 850x600x2000                                                                                                                                                                     |           | 850x1200x2000 |           |           |
|                                    | Module                     | 620x440x131                                                                                                                                                                      |           |               |           |           |
| N.W.(kg)                           | Cabinet                    | 203                                                                                                                                                                              | 235       | 407           | 443       | 513       |
|                                    | Module                     | 33                                                                                                                                                                               |           |               |           |           |
| Executive standard                 |                            | EN/IEC 62040-1,EN/IEC 62040-2,EN/IEC 62040-3;YD/T 2165-2017                                                                                                                      |           |               |           |           |

